

SUNGHO PARK

shpark@dblab.postech.ac.kr, LinkedIn Profile

SUMMARY

I am a Ph.D. student at Data Systems Lab @ Pohang University of Science and Technology (POSTECH), under the supervision of Professor Wook-Shin Han. My research centers on building efficient, reliable, and scalable agentic systems capable of complex reasoning over large-scale multi-modal data (e.g., tables and text). I specialize in designing heterogeneous agentic workflows that integrate SLMs and LLMs to address multi-hop queries. Recently, I have focused on designing automated pipelines for the rigorous evaluation and training of these complex agentic systems.

EDUCATION

Ph.D.	Feb. 2023 - Present	POSTECH, South Korea Graduate School of Artificial Intelligence (Supervisor: Professor Wook-Shin Han)
B.S.	Feb. 2019 - Feb. 2023	POSTECH, South Korea Bachelor of Electrical Engineering Summa Cum Laude (GPA 3.98/4.3; Major GPA 4.10/4.3)
HSD.	Feb. 2017 - Feb. 2019	Chungnam Science High School, South Korea Early graduation

PUBLICATIONS

1. **Park, S.**, Kim, J., and Han, W., “SPARTA: Scalable and Principled Benchmark of Tree-Structured Multi-hop QA over Text and Tables” (*submitted*)
Key Idea: SPARTA is a fully automated SQL-centric pipeline that constructs a tree-structured multi-hop QA benchmark by unifying structured and unstructured evidence from tables and text into a single relational representation.
2. Lee, S., **Park, S.**, and Han, W., “SAFE: Schema-Driven Approximate Distance Join for Efficient Knowledge Graph Querying” (*EMNLP 2025 Main*)
Key Idea: SAFE introduces a schema-driven approximate distance join algorithm that refines noisy LLM-generated query graphs using schema-level constraints and efficiently aligns them with large knowledge graphs, enabling robust and scalable knowledge graph querying.
3. **Park, S.**, Yun, J., Lee, J., and Han, W., “HELIOS: Harmonizing Early Fusion, Late Fusion, and LLM Reasoning for Multi-Granular Table-Text Retrieval” (*ACL 2025 Main*)
Key Idea: HELIOS formulates retrieval as finding a query-relevant subgraph within a bipartite data graph built via early fusion of table segments and passages, and introduces a three-stage pipeline integrating early fusion (SLM), late fusion (SLM), and refinement (LLM).
4. **Park, S.**, Seok, J., Lee, J., Yun, J., and Lee, W., “KDD Cup Meta CRAG 2024 Technical Report: Three-step Question-Answering Framework” (*2024 KDD Cup Workshop for Retrieval Augmented Generation*)
Key Idea: A three-step RAG framework that minimizes unnecessary retrievals by leveraging LLMs’ inherent knowledge and introduces a verification stage to prevent error propagation, improving both accuracy and efficiency in question answering.

EXPERIENCE

Internship @ Oracle Labs	Jul. 2023 - Oct. 2023
Developed an NL2SQL generation model for Oracle Database.	
Internship @ Samsung Electronics’ Mobile Experience Business	Jun. 2022 - Sep. 2022
Developed an image-processing algorithm	

Internship @ AMI Laboratory

Jun. 2021 - Jun. 2022

Developed an end-to-end method for reconstructing an animatable 3D human mesh from a single cropped 2D image of a person.

Internship @ Sellerhub

Jun. 2020 - Sep. 2020

Served as a product manager at Sellerhub, a startup that provides a platform for managing online shopping mall integrations.

PROJECTS

Agentic Workflow for Table-Text Retrieval

Jan. 2024 - Feb. 2025

- Figured out the strengths and weaknesses of both early and late fusion approaches, and developed a graph-based retrieval method that integrates the advantages of both techniques for table-text retrieval.
- Observed that traditional retrieval methods, which rely solely on semantic similarity, struggle with tasks requiring advanced logical inference (such as column-wise aggregation or multi-hop reasoning). Enhanced the retrieval process by incorporating LLM reasoning.
- Analyzed and determined the appropriate granularity for each retrieval stage, optimizing the overall performance. The optimized retrieval method outperforms existing approaches.
- Accepted at ACL 2025 Main

Retrieval-augmented Generation for Open Question Answering

Mar. 2024 - Jun. 2024

- Identified issues with the baseline RAG, including unnecessary retrievals and error propagation from incorrect retrieval results.
- Added two steps to the baseline RAG to address these problems: utilizing latent knowledge within large language models to reduce unnecessary retrievals, and incorporating a verification phase to ensure the factual accuracy of the generated responses.
- Won prizes for the Comparison question in KDD Cup 2024 Meta CRAG Challenge Tasks 1, 2, and 3, as well as for the post-processing question in Task 1.

Development of NL2SQL Generation Model for Oracle Database

Jul. 2023 - Oct. 2023

- Identified that representative NL2SQL benchmarks like Spider follow the SQLite dialect, and more than half of the queries cannot be executed on Oracle databases.
- Built an Oracle-dialect-specific dataset by translating SQLite queries into the Oracle dialect using SQLGlot. SQL queries with SQLite-specific features not supported in Oracle, were excluded from the dev and training sets.
- Improved execution accuracy by 27.4% over the baseline model through various fine-tuning techniques, including hyperparameter tuning, prompt engineering (zero-shot or few-shot), parameter-efficient fine-tuning, and dataset augmentation.

Detecting Anomalies in DB Log Data

Oct. 2023

- Implemented a model for detecting anomalies across multiple DB log datasets.
- Improved the generalization performance of the anomaly transformer (ICLR 2022) by combining scores calculated using Mahalanobis distance, based on multivariate time series statistics, with the scores from the anomaly transformer.

HONORS AND AWARDS

KDD Cup Meta CRAG 2024: Head member of team dRAGonRAnGers

Jul. 2024

- First place in comparison question in Task 1, 2 and 3
- First place in post-processing question in Task 1

Newcomb Lim Ki-Hong Design Challenge Grand Prize	Feb. 2023
Awarded to students of POSTECH EE who presented outstanding research results in a design challenge	
Millitech Military Academy Commander's Award	Feb. 2022
Awarded to a ROND(Benchmarking of Israel's Talpilot)'s novice who showed outstanding research results	
National Science and Technology Scholarship	Aug. 2021
Scholarship awarded to students with outstanding academic ability in the field of science and engineering	

TEACHING EXPERIENCE

Teaching Assistant: AI Application Specialist <i>Samsung Electronics Co.</i>	Aug. 2024 - Aug. 2025
Delivered sessions on language model applications and Retrieval-Augmented Generation (RAG) to software developers.	
Teaching Assistant: Level 4 Data Science Expert Program <i>Samsung Electronics Co.</i>	Jul. 2024
Led hands-on sessions on Retrieval-Augmented Generation (RAG) for data science professionals, with 90% of participants holding Ph.D. degrees.	
Teaching Assistant: Research Project <i>POSTECH</i>	Jun. 2024
Provided mentorship to students Juen Kim and Sohyeon Kim on topics related to Retrieval-Augmented Generation. Juen Kim received the Popularity Award for their project.	
Teaching Assistant: Programming & Problem Solving <i>POSTECH</i>	Feb. 2024
Assisted with course instruction and student support in programming and problem-solving techniques.	

SKILLS

Programming Languages	Python, C, SQL
Python Packages	PyTorch
Tools	PostgreSQL, Docker

AREA OF INTEREST

Retrieval-augmented Generation, Open-domain QA, Multi-modal QA, Information Retrieval, Text2SQL

REFERENCES

Wook-Shin Han	wshan@dblab.postech.ac.kr
Full Professor, Pohang University of Science and Technology (POSTECH), Pohang, South Korea	
Rhicheek Patra	rhicheek.patra@oracle.com
Research Director, Oracle Labs, Zurich, Switzerland	